

Smart Farm-to-Market Supply Chain Optimization Using Power BI

1. Introduction

Agriculture plays a crucial role in the economy, but the farm-to-market supply chain often suffers from inefficiencies such as post-harvest losses, improper storage, long transportation distances, and lack of market visibility. These issues directly affect farmer income and overall supply chain performance.

This project focuses on analyzing the agricultural supply chain using **Power BI** to identify loss patterns, evaluate storage and transportation efficiency, assess market demand, and measure revenue performance. The goal is to support **data-driven decision-making** for stakeholders.

2. Problem Statement

Post-harvest losses, inefficient storage facilities, long-distance transportation, and fluctuating market demand make it difficult for stakeholders to optimize the agricultural supply chain. Due to limited data visibility, decision-makers are unable to identify high-risk crops, inefficient warehouses, and revenue leakage points.

3. Objectives of the Project

- Analyze post-harvest losses across different crops
 - Compare warehouse efficiency (Cold Storage vs Normal Storage)
 - Study the impact of transportation distance on losses
 - Analyze market demand and selling price trends
 - Evaluate total revenue generation by crop and warehouse
 - Assess insurance coverage and supply chain risk
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4. Data Description

Dataset Source: Agricultural Supply Chain & Post-Harvest Loss Dataset (Kaggle)

Key Attributes:

- Crop Type
- Warehouse Type (Cold / Normal)
- Storage Days
- Loss Quantity (kg)

- Loss Reason (Handling, Moisture, Pest, Spoilage)
 - Transport Distance (km)
 - Insurance Flag (Insured / Not Insured)
 - Market Demand Index
 - Selling Price (₹/kg)
 - Total Revenue
 - Date (Month & Year)
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5. Tools and Technologies Used

- **Power BI** – Dashboard creation and visualization
 - **Power Query** – Data cleaning and transformation
 - **DAX** – Calculated measures and KPIs
 - **Excel** – Initial data inspection
 - **SQL** – Data validation and aggregation
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6. Methodology

6.1 Data Preparation

- Removed missing and inconsistent values
- Corrected data types
- Standardized column names
- Created calculated columns and measures

6.2 Data Modeling

- Established relationships between fact and dimension tables
- Optimized data model for performance

6.3 KPI Creation (DAX)

- Total Revenue
 - Total Loss (kg)
 - Average Storage Days
 - Average Selling Price
 - Insured Shipment Percentage
 - Average Transport Distance
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7. Dashboard Design & Analysis

7.1 Executive Overview Dashboard

Provides a high-level summary of total revenue, total losses, average storage duration, insured shipment percentage, and monthly loss trends.

7.2 Loss Analysis Dashboard

Analyzes loss by crop type, loss reason, and warehouse type to identify major contributors to post-harvest losses.

7.3 Market & Revenue Analysis Dashboard

Focuses on revenue generation, selling price trends, market demand index, and monthly revenue patterns.

7.4 Risk, Insurance & Efficiency Analysis Dashboard

Evaluates insurance coverage, transport distance impact on loss, and risk distribution across crops.

7.5 Key Insights & Recommendations Dashboard

Summarizes insights and provides actionable recommendations based on data analysis.

8. Key Insights

- Rice and wheat contribute the highest post-harvest losses
 - Normal storage facilities show higher losses than cold storage
 - Longer transportation distances increase loss risk
 - Nearly half of shipments are uninsured, increasing financial exposure
 - Market demand remains relatively stable, but revenue varies by crop
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9. Recommendations

- Increase usage of cold storage for perishable crops
 - Improve warehouse handling and storage practices
 - Optimize transportation routes to reduce transit losses
 - Expand insurance coverage for high-risk shipments
 - Align pricing strategies with market demand trends
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10. Challenges Faced

- Handling missing and inconsistent data
 - Ensuring accuracy of calculated KPIs
 - Optimizing dashboard performance for large datasets
 - Presenting insights in a business-friendly manner
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11. Future Scope

- Integration of real-time data
 - Inclusion of weather and regional factors
 - Predictive modeling for loss forecasting
 - Cost optimization and profit margin analysis
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12. Conclusion

This project demonstrates how Power BI can be used to transform agricultural supply chain data into actionable insights. The dashboards help stakeholders reduce post-harvest losses, improve storage and transportation efficiency, mitigate risk, and enhance overall revenue performance through data-driven decisions.

Project Type: Capstone / MCA Mini Project

Domain: Data Analytics | Supply Chain Analytics

Tool Used: Power BI